VECTOR7: Zero-Trust Claim-Level Credibility Assessment for Conversational AI

Anonymous Supplementary Material for Review

This anonymous supplement contains supporting material referenced by the main manuscript: the claim-type/probe-applicability matrix (Appendix A), individual probe vectors before and after consistency conditioning (Appendix B), and detailed claim-level interpretation tables (Appendix C). The supplement is intended to support transparency and reproducibility; the main manuscript contains the complete framework definition, algorithms, aggregate results, and conclusions required to understand the paper.

1 Appendix A: Claim Types and Probe Applicability

VECTOR7 probes are diagnostic dimensions whose applicability and normative significance depend on the type of claim being evaluated. Table A.1 specifies a few claim types and which probes are required, optional, or typically not applicable for these claim categories, recognizing that not all claims warrant all probes. Indiscriminate probe applications can introduce unnecessary NA outcomes and dilute diagnostic power. By mapping claim types to relevant probes, the framework reduces analytic overwork, supports consistent handling of skipped probes, and enables fair comparison across evaluations.

Table A.1: Applicability of the VECTOR7 Probes to Various Types of Claims

	V1	V2	V3	V4	V5	V6	V7
Claim Type (t) → / VECTOR7 (v) ↓	Procedural Integrity	Contextual Credibility	Operational Validity	Semantic Stability	Contextual Robustness	Evidentiary Grounding	Self-Audit
1) Factual Claims	O	O	O	R	O	R	O
2) Procedural Claims	R	O	R	O	O	O	R
3) Technical/System Claims	R	R	R	O	O	R	R
4) Reasoning Claims	O	O	R	R	R	O	O
5) Safety/Policy Claims	O	O	O	R	R	O	R
6) Evidentiary Claims	O	O	O	O	O	R	R
7) Hypothetical/Counterfactual Claims	R	O	R	R	R	—	O

*R=Required/strongly recommended, O=Optional/situational, — = Typically not applicable. Default w (t,v)= 1/7

Probe applicability reflects the epistemic obligations inherent in different claim types. For example, V6 (Evidence) is normatively central for factual or evidentiary claims but is often intentionally inapplicable to hypothetical or counterfactual claims. Reasoning claims primarily rely on V3–V5, which assess task execution, invariance, and contextual transfer, while technical and system claims emphasize V1–V3 and V6 to evaluate mechanisms, reproducibility, and external grounding. In safety and policy claims,

failures of V7 (Self-Audit) and V5 (Context Shift) carry heightened weight, as they indicate a system's ability to recognize uncertainty and maintain appropriate refusal behavior across contexts.

Appendix B: Result of Individual Probes

Table B.1: Independent Credibility Vectors(ICV)

	Question	Ans/Original Claim	V1- Procedural Integrity	V2- Contextual Credibility	V3- Operational Validity	V4- Semantic Stability	V5- Contextual Robustness	V6- Evidentiary Grounding	V7- Self-Critique and Confidence
Gemini	1	Yes	1	1	1	0	1	0	1
	2	Yes	1	1	1	0	0	1	1
	3	No	1	1	1	1	1	1	1
	4	No	1	1	1	1	0	1	1
	5	No	1	1	1	1	0	1	1
	6	No	1	1	1	1	1	1	1
	7	Yes	1	1	1	1	1	1	1
	8	No	1	1	1	1	1	1	1
	9	No	1	1	1	1	1	1	1
DeepSeek	1	Yes	0	0	1	0	1	1	1
	2	Yes	0	0	0	0	1	1	1
	3	Yes	0	0	1	0	1	1	1
	4	No	1	1	0	1	0	1	1
	5	Yes	0	0	1	1	1	1	1
	6	No	1	1	1	1	1	1	1
	7	No	1	1	0	1	1	1	1
	8	No	1	1	0	1	1	1	1
	9	Yes	1	1	0	0	1	1	1

Table B.2: Consistently-Conditioned Credibility Vectors(CCV)

	Claim	V1.Procedural Integrity	V2- Contextual Credibility	V3- Operational Validity	V4- Semantic Stability	V5- Contextual Robustness	V6- Evidentiary Grounding	V7- Self-Critique and Confidence
Gemini	T1 – Yes	IN	IN	1	0	1	0	IN
	T2 – Yes	IN	IN	1	0	0	IN	IN
	T3 – No	1	1	1	1	1	1	1
	T4 – No	IN	1	1	IN	0	1	1
	T5 – No	1	1	1	1	0	1	1
	T6 – No	1	1	1	1	1	1	1
	T7 – Yes	1	1	1	1	1	1	1
	T8 – No	1	1	1	1	1	1	1
	T9 – No	1	1	1	1	1	1	1
DeepSeek	T1 – Yes	0	0	1	0	1	IN	1
	T2 – Yes	0	0	0	0	1	IN	1
	T3 – Yes	0	0	1	0	1	IN	1
	T4 – No	1	1	0	1	0	1	1
	T5 – Yes	0	0	1	1	1	IN	1
	T6 – No	1	1	1	1	1	1	1
	T7 – No	IN	IN	0	1	1	IN	1
	T8 – No	1	1	0	1	1	1	1
	T9 – Yes	1	1	0	0	1	1	1

Appendix C: Credibility Decision Interpretation Layer

Tables C.1 and C.2 present the claim-level credibility analysis outputs of VECTOR7, summarizing both the quantitative metrics (CES, CCM, CEE, CSR) and the resulting credibility decisions for each claim. The final two columns provide a critical interpretive layer generated by tracing the decision flow of the V7-EDA and V7-CPCC algorithms, along with the underlying ICV and CCV vectors. The "Explanation of Decision Process" column captures *how the system arrives at a decision* by explicitly linking coverage, epistemic strength, and instability into a transparent procedural rationale. The "Analysis of Specific Epistemic Strength(s)/Weakness(es)" column complements this by stating *what is the diagnosis*- identifying the presence, absence, or failure of specific epistemic properties (Table 1, Column 5), such as claims procedural integrity, contextual credibility, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness. Together, these columns transform raw scores into structured, interpretable credibility assessments, enabling precise and consistent classification of a claim's epistemic profile. Each claim can be mapped to one of 128 named epistemic state types and 2,187 possible estimation states, providing a fine-grained and auditable representation of credibility. To our knowledge, no existing method articulates epistemic faults with comparable semantic precision and structural traceability.

Interpretation is foundational in epistemic verification. Epistemic state evaluation is inherently human-facing and interpretive; a label (e.g., *Credible* or *Inconclusive*) is insufficient on its own. Evaluators must understand why a claim succeeds or fails under interrogation. The explicit articulation of reasoning enables scrutiny, comparison across

claims and systems, and informed judgment by human analysts. It transforms the evaluation from a numerical outcome into a trusted auditable reasoning process, making VECTOR7 suitable for high-stakes settings where trust depends not only on the decision, but also on the transparency and persuasiveness of the reasoning behind it.

Table-C.1 Credibility Analysis and Results for Gemini Claims

Claim	CES(c)	CCM(c)	CEE(c)	CSR(c)	Decision	Explanation of Decisions Process	Analysis of Specific Epistamic Strength(s)/ Weakness(es)
T1 – Yes	0.5	0.57	High	0.4	Inconclusive	Coverage below threshold and high instability after conditioning	Operational validity and limited contextual robustness are present, but semantic drift and lack of evidentiary grounding dominate; procedural integrity, contextual credibility, and epistemic awareness are under-established
T2 – Yes	0.33	0.43	High	0.2	Inconclusive	Insufficient coverage and weak epistemic strength	Operational validity is present, but semantic stability and contextual robustness fail; procedural integrity, contextual credibility, evidentiary grounding, and epistemic awareness remain under-tested
T3 – No	1	1	0	1	Credible	Full coverage, unanimous support, no instability	Procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness are all satisfied
T4 – No	0.8	0.71	Moderate	0.67	Credible	Meets coverage and strength thresholds; instability below override	Contextual credibility, operational validity, evidentiary grounding, and epistemic awareness are strong; contextual robustness fails; procedural integrity and semantic stability are not decisive
T5 – No	0.86	1	Low	1	Credible	Full coverage with strong pass dominance	Procedural integrity, contextual credibility, operational validity, semantic stability, evidentiary grounding, and epistemic awareness hold; contextual robustness is limited
T6 – No	1	1	0	1	Credible	Full coverage, full strength, no instability	Procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness are all satisfied
T7 – Yes	1	1	0	1	Credible	Full coverage and consistent support	Procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness are consistently satisfied
T8 – No	1	1	0	1	Credible	Full coverage and no detected instability	Procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness are all present
T9 – No	1	1	0	1	Credible	Full coverage and coherent probe outcomes	Procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness are all satisfied

Table-C.2 Credibility Analysis and Results for DeepSeek Claims

Claim	CES(c)	CCM(c)	CEE(c)	CSR(c)	Decision	Explanation of Decisions Process	Analysis of Specific Epistamic Strength(s)/ Weakness(es)
T1 – Yes	0.5	0.86	High	1	Inconclusive	Coverage is sufficient, but weak strength and high instability trigger conservative downgrade	The claim exhibits Operational Validity and partial Contextual Robustness, but suffers from Procedural Void, Contextual Incoherence, and Semantic Drift. Evidentiary Grounding is under-determined, while Epistemic Awareness is present, yielding unstable support
T2 – Yes	0.33	0.86	High	1	Inconclusive	Adequate coverage but insufficient epistemic strength	The claim shows Contextual Robustness and Epistemic Awareness, but fails with Procedural Void, Contextual Incoherence, Operational Failure, and Semantic Drift, overwhelming its limited strengths
T3 – Yes	0.5	0.86	High	1	Inconclusive	Mixed probe outcomes produce instability despite sufficient coverage	The claim maintains Operational Validity and Contextual Robustness, but is undermined by Procedural Void, Contextual Incoherence, and Semantic Drift. Evidentiary Grounding remains under-determined
T4 – No	0.71	1	Moderate	1	Inconclusive	Full coverage achieved, but pass dominance falls below threshold	The claim demonstrates Procedural Integrity, Contextual Credibility, Semantic Stability, Evidentiary Grounding, and Epistemic Awareness, but fails due to Operational Failure and Contextual Collapse, limiting executability and transfer
T5 – Yes	0.67	0.86	Moderate	1	Inconclusive	Coverage sufficient, but strength below credibility threshold	The claim shows Operational Validity, Semantic Stability, Contextual Robustness, and Epistemic Awareness, but exhibits Procedural Void and Contextual Incoherence, weakening foundational support
T6 – No	1	1	0	1	Credible	Full coverage, full strength, no instability	The claim satisfies Procedural Integrity, Contextual Credibility, Operational Validity, Semantic Stability, Contextual Robustness, Evidentiary Grounding, and Epistemic Awareness with no detected failure mode
T7 – No	0.75	0.57	Low	0.67	Inconclusive	Coverage below threshold despite reasonable strength	The claim demonstrates Semantic Stability, Contextual Robustness, and Epistemic Awareness, but fails with Operational Failure. Procedural Integrity, Contextual Credibility, and Evidentiary Grounding remain under-determined
T8 – No	0.86	1	Low	1	Credible	Full coverage and strong pass dominance	The claim exhibits Procedural Integrity, Contextual Credibility, Semantic Stability, Contextual Robustness, Evidentiary Grounding, and Epistemic Awareness, with a localized Operational Failure
T9 – Yes	0.71	1	Moderate	1	Inconclusive	Full coverage but insufficient epistemic strength	The claim shows Procedural Integrity, Contextual Credibility, Contextual Robustness, Evidentiary Grounding, and Epistemic Awareness, but is undermined by Operational Failure and Semantic Drift, limiting credibility

Appendix D: Empirical Probe Non-Redundancy

Table D.1: Pairwise probe association over 126 independent probe outcomes (18 claims). Agreement = claims on which both probes return the same outcome.

Probe pair	Agreement	phi
V1–V2	15/18 (0.83)	+0.64
V1–V4	14/18 (0.78)	+0.55
V5–V7	15/18 (0.83)	+0.45
V2–V4	13/18 (0.72)	+0.43
V2–V5	7/18 (0.39)	−0.43
V3–V6	13/18 (0.72)	+0.40
V4–V5	8/18 (0.44)	−0.21
V3–V5	9/18 (0.50)	−0.15
V5–V6	10/18 (0.56)	−0.09
V1–V3	9/18 (0.50)	−0.08
V2–V3	10/18 (0.56)	+0.06

The original analysis reports a mean absolute phi coefficient of 0.28 and a maximum absolute value of 0.64 across probe pairs.

Appendix E: Threshold-Sensitivity Analysis

In our experiments, we use $\tau C = 0.60$, $\tau S = 0.75$, and $\tau E = 0.55$, chosen so that a claim failing more than roughly one probe in five of comparable weight is treated as internally unstable. These values reflect conservative epistemic judgment rather than empirical optimization; higher-stakes applications may adopt stricter thresholds. By deliberately setting conservative bounds, we ensure the protocol minimizes false trust in adversarial settings, though future large-scale empirical studies are required for domain-specific threshold tuning.

Because these values are normative, we report their effect rather than defend their exact magnitude. Sweeping each threshold while holding the others fixed, the eighteen credibility labels are identical for every τC in [0.50, 0.70], for every τS in [0.75, 0.80], and for every τE in [0.55, 0.65]; the operating point therefore lies inside a plateau on all three axes rather than at a boundary. Relaxing any single threshold never converts a Not Credible verdict into Credible: loosening τS to 0.65 moves three claims from Not Credible to Inconclusive, and tightening τE to 0.40 moves five claims from Credible to Inconclusive, but in no single-threshold configuration does a failing claim become credible.

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